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Rheumatoid and Osteoarthritis

Peter Miller, Arthritis in Health Claims, February 2011,
Health Claims Forum

Case 1: Rheumatoid arthritis

- 43 year old housewife
- Seronegative rheumatoid arthritis of right hand for 3 years
- Synovitis of her MCP joints and right dorsal tenosynovitis
- Methotrexate (hasn't worked), Arcoxia (Etoricoxib), Avara (Leflunomide)
- Generalised physical weakness, in particular right hand
- Pain in right hand with swelling, nodules and arthritic changes – wakes her regularly
- Increased fatigue and lack of energy since onset
- Unable to handle housework, pick up children or play sports with them
- Claim declined as not 3 joint groups

Policy Rheumatoid Arthritis CI definition

- Widespread joint destruction with major clinical deformity of three or more of the following joint areas: hands, wrists, elbows, cervical spine, knees, ankles, metatarsophalangeal joints in the feet. The condition should be such that it results in a permanent inability to perform any three (3) of the following Activities of Daily Living (ADLs):
 - Dressing – the ability to put on and take off clothing without assistance
 - Mobility – the ability to move from room to room without physical assistance
 - Transfer – the ability to get in and out of bed or a chair without assistance
 - Continence – the ability to control bowel and bladder function
 - Feeding – the ability to get food from a plate into the mouth without assistance
 - Bathing and showering – the ability to bathe and shower without assistance

Medical definition of Rheumatoid Arthritis

- Rheumatoid arthritis (RA) is a **chronic, systemic**, inflammatory disorder of unknown etiology that primarily involves joints. The arthritis is symmetrical, may be remitting, but if uncontrolled may lead to destruction of joints due to erosion of cartilage and bone which leads to deformity. The disease usually progresses from the periphery to more proximal joints, and in patients who do not fully respond to treatment, results in significant locomotor disability within 10 to 20 years.



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Clinical features of Rheumatoid Arthritis

- Morning stiffness for at least one hour and present for at least six weeks
- Swelling of three or more joints for at least six weeks
- Swelling of wrist, metacarpophalangeal (MCP), or proximal interphalangeal (PIP) joints for at least six weeks, often symmetrical
- Hand x-ray changes typical of RA that must include erosions or unequivocal bony decalcification
- Rheumatoid subcutaneous nodules
- Rheumatoid factors

Clinical course of Rheumatoid Arthritis

- Fluctuation of disease activity over periods lasting weeks to months
- Disease activity may not abate in about 10 to 20 percent of cases
- Disease activity needs to be distinguished from destructive activity leading to permanently damaged joints
 - disease activity is variable
 - structural damage is cumulative and irreversible
- Remission is very rare in patients with a well established diagnosis of RA without use of disease-modifying anti-rheumatic drugs (DMARDs)

Physical findings in Rheumatoid Arthritis

■ Early findings:

- pain and swelling of the affected joints
- local tenderness and painful movement
- synovial thickening of joints and tendons
- reduced grip strength if hands involved

■ Late findings:

- ulnar deviation of fingers and toes
- swan neck or Boutonniere deformities of fingers
- bone erosions
- joint destruction

Synovial thickening of the metacarpophalangeal joint



Bilateral swelling of the MCP joints is evident in this patient with rheumatoid arthritis. Note also the mild swan neck deformities present in several fingers, particularly the left middle and fifth fingers.

Courtesy of Patrick J Venables, MD.



Swelling of the metacarpophalangeal joints of the right hand



Swelling of the MCP joints, moderate MCP flexion, and swan neck deformities are evident in this patient with rheumatoid arthritis.

Courtesy of Patrick J Venables, MD.



RA of the elbow



Lateral view of the elbow in a patient with rheumatoid arthritis (RA) reveals soft tissue swelling and osteopenia with destruction of the elbow joint (arrows). There are also secondary proliferative bony changes, which have arisen due to joint space destruction.

Courtesy of Jonathan Kruskal, MD.



Investigations in Rheumatoid Arthritis

- Acute phase reactants - reflect the degree of synovial inflammation
 - C-reactive protein (CRP) and erythrocyte sedimentation rate (ESR)
- Haemoglobin – usually less than normal, but > 9.0 g/dl, unless GI bleeding
- Rheumatoid factor (RF) titres – only useful in initial diagnosis and not for following disease activity
- Anti-citrullinated peptide antibody (ACPA) titres – as sensitive and more specific than RF for diagnosis of RA. Better predictor for erosive joint disease than RF in early disease
- Radiographs – hands and wrists and both feet early in disease to set baseline

Rheumatoid factor

- Rheumatoid factors (RF) are autoantibodies found in 75 to 80% of RA patients
- RF also occurs in connective tissue diseases, SLE and primary Sjögren's syndrome, and with certain infections - malaria, rubella, hepatitis C, and after vaccinations
- Seropositive RA often associated with more aggressive joint disease, and is more commonly complicated by extra-articular manifestations than seronegative RA
- Rheumatoid nodules and vasculitis occur almost exclusively in seropositive patients, associated with increased mortality
- Radiographic progression may be more rapid among patients with a positive RF at initial evaluation
- Study of 135 women with early RA found that persistently positive RF had more erosions, nodules, extra-articular disease, functional disability and disease activity than seronegative, or intermittently seronegative individuals over 6 years follow up

Severity of RA

- Mild disease —meet criteria for RA and typically <6 inflamed joints, no extra-articular disease, and no erosions or cartilage loss
- Severe disease —typically >20 inflamed joints, elevated ESR and/or CRP, plus:
 - Anaemia and/or hypoalbuminaemia
 - RF positivity (often in high titer) and/or anti-CCP antibodies
 - Radiographs with rapid appearance of bony erosions and loss of cartilage
 - Extra-articular disease
- Moderate disease —neither mild nor severe disease, typically 6 -20 inflamed joints and plus combination of the following clinical features:
 - Absence of extra-articular disease (most commonly)
 - Elevation in ESR and/or CRP and positive RF and/or anti-CCP antibodies
 - Inflammation plain radiography, such as osteopenia and/or periarticular swelling; in addition, minimal joint space narrowing and small peripheral erosions may be observed.
 - Often progress to severe disease after a period of inadequately controlled synovitis.

Extra-articular disease in Rheumatoid Arthritis

- Cytokines that cause synovitis also responsible for pathology in extra-articular tissues
- Osteopenia – disease, steroid treatment and immobility
- Muscle weakness – synovitis, myositis, vasculitis, steroids
- Skin disease – rheumatoid nodule, skin ulcers
- Eye involvement – episcleritis, scleritis
- Lung disease – pleuritis , effusion, interstitial fibrosis, pulmonary nodules, and bronchiolitis obliterans and organizing pneumonia
- Cardiac disease – pericarditis, myocarditis, CAD
- Peripheral vascular disease – vasculitis, PAD, stroke
- Blood – anaemia, Felty's syndrome

Treatment of Rheumatoid Arthritis - general

- Destruction of affected joints occurs early in RA – essential that established disease should be treated with disease modifying anti-rheumatic drugs (DMARDs) as soon as possible
- Severe active disease and/or poor prognostic signs should be treated more aggressively than milder disease
- Escalate intensity of treatment until synovitis and inflammation diminishes or disappears, or until drug-induced side effects become intolerable.
- Patients with RA are a heterogeneous group, requiring different treatment approaches
 - very mild and undifferentiated disease
 - early, established, and severe disease
 - end-stage disease, with little inflammation but severe functional disabilities

Treatment of Rheumatoid Arthritis - drugs

- Analgesics —acetaminophen (paracetamol), propoxyphene, tramadol, and more potent opioids (e.g., oxycodone, hydrocodone)
- Non-steroidal anti-inflammatory drugs (NSAIDs) have both analgesic and anti-inflammatory properties but do not alter disease outcomes – celecoxib, naproxen, ibuprofen, diclofenac
- Glucocorticoids —prednisone or prednisolone suppress inflammation; oral, IV, or intra-articular . Low doses up to 7.5 mg/day relatively safe, but higher doses associated with significant side effects
- Disease-modifying anti-rheumatic drugs (DMARDs) reduce or prevent joint damage, preserve joint integrity and function
 - Nonbiologic: hydroxychloroquine, sulfasalazine, methotrexate, leflunomide, and minocycline
 - Biologic: TNF- α inhibitors - etanercept, infliximab, adalimumab, golimumab, and certolizumab pegol. Others: anakinra, tocilizumab, abatacept, rituximab

Prognosis of Rheumatoid Arthritis

- Long clinical remissions —Almost 10 percent of patients have prolonged clinical remissions that may be associated with disappearance of rheumatoid factor positivity. Some of these patients have an occasional flare, but the prognosis for good function is excellent.
- Intermittent disease —Approximately 15 to 30 percent of patients have an intermittent course characterized by partial to complete remissions, without need for continuous therapy. The remissions last up to one year, and relapses are often marked by involvement of additional joints.
- Progressive disease —The majority of patients have progressive disease that can lead to joint destruction and disability.

ABI proposed Rheumatoid Arthritis CI definition, September 2009

- **Chronic severe rheumatoid arthritis** – resulting in a loss of the ability to do specified physical activities
- A definite diagnosis by a consultant rheumatologist of chronic rheumatoid arthritis as evidenced by widespread joint destruction with major clinical deformity.
- In addition the claimant must permanently satisfy three of the four following criteria:
- **Bending** – The inability to bend or kneel to pick up something from the floor and stand up again and the inability to get into and out of a standard saloon car.
- **Dexterity** – The inability to use hands and fingers to pick up and manipulate small objects such as cutlery, including being unable to write using a pen or pencil.
- **Lifting** – The inability to lift, carry or otherwise move everyday objects by hand. Everyday objects include a kettle of water, a bag of shopping and an overnight bag or briefcase.
- **Mobility** – The inability to walk a distance of 200 metres on flat ground, even with the aid of a walking stick if prescribed by a treating practitioner, and without having to rest.

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Limitations due to Rheumatoid arthritis

- Patients with RA form a heterogeneous group
 - very mild and undifferentiated disease
 - early, established, and severe disease
 - end-stage disease, with little inflammation but severe functional disabilities
- Fatigue, weakness and morning-stiffness
- Joint pain, reduced ROM, weakness from active arthritis and synovitis
 - Hands – loss of dexterity limiting writing and keyboard use
 - Feet and ankles – difficulty in standing and walking
- Reduced ROM, weakness, pain from joint deformity and destruction which is permanent
- Symptoms related to extra-articular disease
- Depression

Case 2: Osteoarthritis

- 52 year old male anaesthetist and critical care team leader
- Previously fit active person – competitive squash and swimming and farming
- 2001 left knee pain, osteoarthritis, 2004 arthroscopy and meniscectomy and 2008 total left knee replacement
- Walking limited to 10-15 minutes and limited standing
- Right knee now involved
- Pain and limitation in hands, CMC and MCP joints bilaterally, neck pain and crepitus – can't pick up heavy patient files
- Coccydynia (treated by MUA) limits sitting
- Radiographs show CMC and MCP osteoarthritis, osteophytes and subluxation
- Unable to perform functions of anaesthetist and feels hazardous to patients

Medical definition of Osteoarthritis

- Osteoarthritis was previously thought to be a normal consequence of aging, thereby leading to the term degenerative joint disease
- Osteoarthritis results from articular cartilage failure induced by a complex interplay of multiple factors, including joint integrity, genetic predisposition, local inflammation, mechanical forces, and cellular and biochemical processes
- The process involves interactive degradation and repair processes of cartilage, bone, and synovium

Types of Osteoarthritis

- Idiopathic osteoarthritis can be categorized into localized or generalized.
 - Localized OA most commonly affects the hands, feet, knee, hip, and spine.
 - Generalized OA consists of involvement of three or more joint sites.
- Secondary osteoarthritis —Specific conditions may cause or enhance the risk of developing osteoarthritis:
 - Trauma
 - Congenital or developmental disorders
 - Calcium crystal deposition disease
 - Other bone and joint disorders including rheumatoid arthritis, gouty arthritis, septic arthritis, and Paget disease of bone
 - Other diseases such as diabetes mellitus, acromegaly, hypothyroidism, neuropathic (Charcot) arthropathy, and frostbite

Risk factors and causes of Osteoarthritis

- Age
- Female versus male sex
- Obesity
- Lack of osteoporosis
- Occupation
- Sports activities
- Previous injury
- Muscle weakness
- Proprioceptive deficits
- Genetic elements
- Acromegaly
- Calcium crystal deposition disease

Diagnosis of Osteoarthritis

- The diagnosis of osteoarthritis is complicated
 - lack of specific physical or laboratory findings and discrepancies between symptoms and the results of radiographic examinations
- Usually diagnosed by an overall clinical impression based upon the patient's age and history, findings on physical examination, and radiographic findings
- Standardization of the definition, classification, and diagnosis of osteoarthritis has been formulated for affected anatomic areas - including the knee, hand, and hip

Clinical manifestations of Osteoarthritis

- Age of onset - usually after age 40
- Commonly affected joints: cervical and lumbar spine, 1st CMC, proximal and distal IP joints, hip, knee, subtalar joint, 1st MTP joint
- Uncommonly affected joints: shoulder, wrist, elbow , MCP joint
- Symptoms: pain, stiffness and gelling
- Findings on physical examination: crepitus, bony enlargement, decreased range of motion, malalignment, tenderness to palpation
- Synovial fluid analysis: clear fluid, WBC <2000/mm³, normal viscosity
- X-ray features: joint space narrowing, subchondral sclerosis, marginal osteophytes, subchondral cysts
- Patterns of presentation: monoarticular in young adult, pauciarticular, large-joint in middle age, polyarticular generalized, rapidly progressive, secondary to trauma, congenital abnormality, or systemic disease

Osteoarthritis of the distal interphalangeal (DIP) joints



This plain film demonstrates complete loss of the articular cartilage at all four DIP joints, large osteophytes, and ankylosis of the DIP joint of the middle finger.

Courtesy of Bruce C Anderson, MD.



Osteoarthritis of the knee



This preoperative radiograph (x-ray) shows severe loss of cartilage affecting the medial compartment of the knee.

Courtesy of Mark L Robbins, MD, MPH.



Hip osteoarthritis

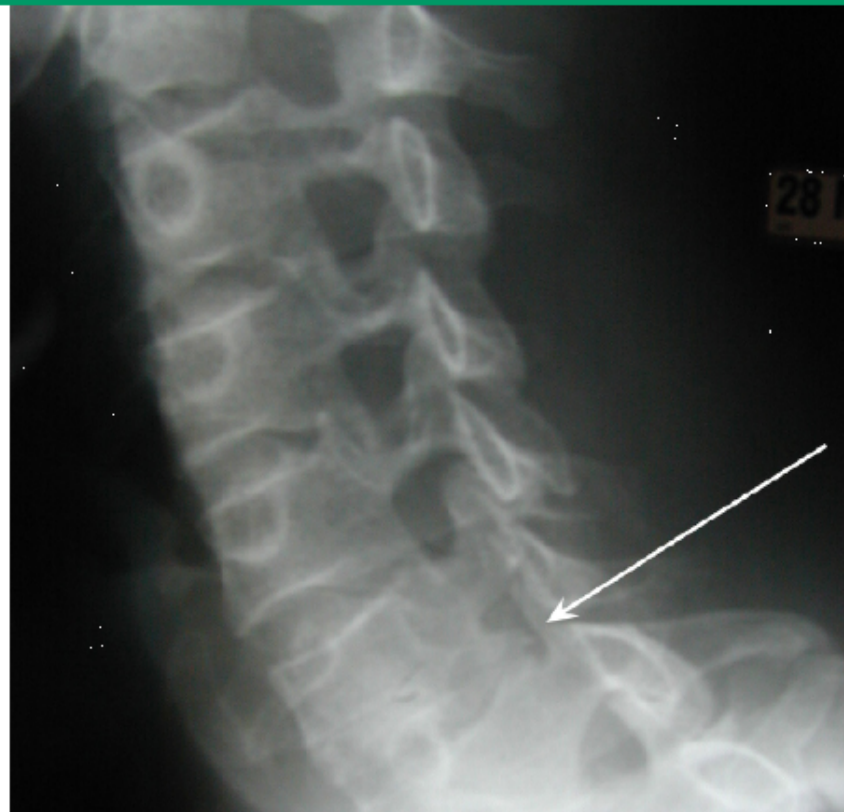


Plain radiograph of the right hip joint demonstrates marked joint space narrowing, sclerosis about opposing joint space margins, and periarticular osteophyte formation. These features are all characteristic of osteoarthritis.

Courtesy of Jonathan Kruskal, MD.



Oblique neck film for foraminal encroachment



Oblique views are primarily used to determine the role of foraminal encroachment. This 54-year-old patient presented with a gradual onset of numbness and tingling of the middle fingers of the right hand. Symptoms waxed and waned over a period of several months. The triceps reflex and muscle strength were normal. The oblique film demonstrates C6-7 foraminal encroachment.

Courtesy of Bruce C Anderson, MD.



Treatment of Osteoarthritis

- Rest when painful and inflamed
- Exercise to improve mobility and strength
- Analgesics – acetaminophen, opioids
- NSAIDS – oral, topical
- Intra-articular glucocorticoids
- Surgery – partial or total joint replacement

Prognosis of Osteoarthritis

- Prognosis : variable, generally slowly progressive
- Hand OA - older age appears to be the strongest risk factor for progression
- Hip OA - risk factors for progression to total hip replacement include female gender, night pain, and lower baseline functional capacity
- Knee OA - a higher BMI and polyarticular arthritis predict X-ray deterioration. Varus or valgus deformity of the knee results in later X-ray worsening
- Women with unilateral knee OA - 34% developed contralateral disease by X-ray criteria at 24 months and risk is 5-fold higher between highest and lowest BMI terciles
- Worsening disability related to coping styles, e.g. avoidance of activity due to pain may lead to muscle weakness that may reduce joint stability. Exercise can help to prevent such loss of strength and decrease disability, yet exercise can cause pain

Limitations due to Osteoarthritis

- Depends on joints involved – hands, hips, knees, feet and cervical and lumbar spine
- Hands – weakness, loss of dexterity, pain
 - dressing, writing, keyboard, manual labour
- Hips, knees and feet – pain, weakness
 - walking, standing, bending down, squatting, kneeling
- Cervical and lumbar spine – pain, weakness
 - stiff neck (driving), arm and leg nerve entrapment (pain and weakness), bending, walking, sitting, lifting, carrying

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- Pain and limitation hands, CMC and MCP joints 2-3 bilaterally, neck pain and crepitus – unable to hold heavy file
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- Radiographs show CMC and MCP osteoarthritis, osteophytes and subluxation
- Unable to perform functions of anaesthetist and colleagues feel hazardous to patients



New Classification Criteria for Rheumatoid Arthritis

The goal is early identification of patients who can benefit from intervention.

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Questions?

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